

Project 5

In the following exercise, we will perform bootstrap and cross validation to find statistical values for two datasets. Perform the following analyses by starting a new notebook.

Part A

First, we will use the Boston dataset to try running our own implementation of bootstrap.

1. Based on this dataset, provide an estimate for $\hat{\mu}$, the population mean of 'medv'. Name this variable mu_hat in Python and **enter your answer**.
2. Provide an estimate of the standard error of $\hat{\mu}$. Name this variable se_mu_hat and **enter your answer**. Interpret this result in your Jupyter notebook.

Hint: We can compute the standard error of the sample mean by dividing the sample standard deviation by the square root of the number of observations. When computing the sample standard deviation, one must consider Bessel's correction factor from statistics (more information on that [here](#)); therefore, the appropriate degrees of freedom should be 1. Verify that the standard error you calculated matches the one from *scipy.stats.sem*.

3. Now estimate the standard error of by implementing your own bootstrap method. Name your final answer se_bootstrap and **enter your answer**. How does this compare to your answer from Part A Question 2? *Hint:* Generate $M = n$ artificial datasets, each with $N = n$ bootstrap samples. Use `np.random.seed(42)` before sampling so that your outputs stay consistent.
4. Based on your bootstrap estimate from Part A Question 3, provide a 95% confidence interval for the mean of 'medv' and **enter your answer**. *Hint:* You can approximate a 95% confidence interval using the formula $[\hat{\mu} \pm 2 * SE(\hat{\mu})]$
5. Based on this data set, provide the population median value of 'medv', $\widehat{median}$. Name this variable median_hat and **enter your answer**.

6. We now would like to estimate the standard error of the $\widehat{median}$. Unfortunately, there is no simple formula for computing the standard error of the median. Instead, estimate the standard error of the median using bootstrap. Name this variable `se_bootstrap_median` and **enter your answer**.
7. **Answer the Question: Is the standard error of the median greater or less than the standard error of the mean?** Does this make intuitive sense? Comment on your other findings in your Jupyter notebook.

Part B

Next, we will use the Default dataset to predict the probability of default using income and balance. In doing so, we also want to estimate the test error of the logistic regression model described in that section using cross validation.

1. Fit a logistic regression model that use all the data from the predictors 'income' and 'balance' to predict 'default'. *Hint:* Use Stats Models to fit logistic regression. Review the previous recitations for review.
2. Using the validation set approach, estimate the test error of this model. In order to do this, you must perform the following steps:
 - (a) Split the sample set into a random training set and a random validation set. Use a test size of 20%. *Hint:* Use `sklearn.model_selection.train_test_split` and a `random_state` of 42.
 - (b) Fit a logistic regression with multiple variables model using only the training observations.
 - (c) Obtain a prediction of 'default' for each individual in the validation set by computing the probability of default for that individual and classifying the individual to the default category if the posterior probability is greater than 0.5 (a Bayesian classifier).
 - (d) Compute the misclassification rate, which is the fraction of the observations in the validation set that are misclassified. Name this variable `mis_rate` and **enter your answer**. *Hint:* Compare the predictions with the test set.
3. Repeat the process in Part B Question 2 three times, using three different splits of the observations into a training set and a validation set. Use test sizes of 10%, 30%, and 50%. Name the misclassification rates `mis_rate_10`, `mis_rate_30`, and `mis_rate50`, respectively. Comment on the results obtained and determine the key takeaway. **Enter your answers**.
4. Using KFold cross validation with five folds across 5 trials, calculate the average misclassification rate. Name this `mis_rate_kfold` and enter **your answer**. In your notebook, briefly explain the KFold process. *Hint:* Because Stats Models does not have its own cross validation libraries, you may have to use the following packages:
`sklearn.linear_model.LogisticRegression`
`sklearn.model_selection.cross_val_score`
`sklearn.model_selection.KFold` (specify a `random_state` equaling the trial number).
5. Using bootstrap on the complete dataset, compute estimates for the standard errors of the income and balance logistic regression coefficients. In order to do this, you should perform the following steps:

- (a) Write a function that takes as input the Default data set as well as an index of the observations. The method should be called `boot_sample(data, index)`. The method should return the coefficient estimates for income and balance in the logistic regression model.
- (b) Use this function to estimate the standard errors of the logistic regression coefficients for income and balance using 100 different bootstrap samples. *Hint:* You can first sample the dataset's indices and then use those sampled indices as inputs for your defined method.
- (c) Comment on the estimated standard errors obtained. What are the confidence intervals of the logistic regression coefficients? **Enter your answers.**